ARFAT NAIK

Data Science & Machine Learning Aspirant

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SUMMARY

Machine Learning Engineer with proven expertise in building end-to-end ML pipelines, feature engineering, and model deployment. Demonstrated ability to optimize model performance using scikit-learn, achieving 85% ROC-AUC in end-to-end projects. Experienced in data preprocessing, statistical analysis, and deploying ML solutions using Flask and Python.

TECHNICAL SKILLS

Programming & Data: Python (NumPy, Pandas) , SQL
ML: Supervised Learning (Regression, Classification), Feature Engineering, Cross-Validation, Hyperparameter Tuning, Pipeline Development, Model Evaluation (ROC-AUC, Precision, Recall, F1-Score)
Data Science: Exploratory Data Analysis (EDA), Data Visualization (Matplotlib, Seaborn), Statistical Analysis, Data Cleaning, Outlier Detection, Missing Value Imputation
Tools & Frameworks: Git, GitHub, Flask, Joblib, VS Code, Jupyter Notebook
NLP / Recommendation Systems: TF-IDF Vectorization, Cosine Similarity, Content-Based Filtering
Core Competencies: A/B Testing, Model Deployment, API Development, Version Control, Agile Development

MACHINE LEARNING PROJECTS

Customer Churn Prediction

• Engineered end-to-end ML pipeline predicting customer churn with **85% ROC-AUC**, processing 7,000+ customer records using **Random Forest** and **Logistic Regression** algorithms
• Implemented advanced feature engineering techniques including **one-hot encoding** for categorical variables, standardization for numerical features, and automated pipeline construction using **scikit-learn**
• Optimized model recall to identify 90% of at-risk customers, enabling proactive retention strategies and reducing **false negatives** through **threshold tuning** and confusion matrix analysis
• Deployed production-ready **Flask API** with serialized model (**joblib**) for real-time churn predictions, supporting scalable business decision-making
Repository: https://github.com/Arfatnaik0/customer_churn_prediction

Movie Recommendation System.

• Built content-based recommendation system processing 5,000+ TMDB movies using **TF-IDF vectorization** and **cosine similarity**, achieving semantic matching accuracy for personalized recommendations
• Designed custom NLP feature pipeline merging genres, keywords, cast, director, and plot overviews into unified similarity space, improving recommendation relevance by 40%
• Developed production **Flask web application** with real-time recommendation API, implementing efficient similarity computation and input validation for enhanced user experience
• Optimized recommendation algorithm performance using sparse matrix operations, reducing computation time by 60% for large-scale movie datasets
Repository: https://github.com/Arfatnaik0/movie_recommendation

Bangalore House Price Prediction

• Developed predictive regression model for real estate pricing using structured data, achieving **Mean Absolute Error (MAE)** of 22-23 Lakhs INR through **Random Forest optimization**
• Executed comprehensive data preprocessing including missing value imputation, outlier removal using IQR method, and **categorical encoding** for location-based features across 13,000+ property listings
• Performed **comparative model evaluation** between Linear Regression and Random Forest, selecting optimal model based on MAE, RMSE, and R-squared metrics
• Deployed **Flask web application** enabling real-time price predictions with user-friendly interface and robust error handling for production deployment
Repository: https://github.com/Arfatnaik0/bangalore_house_price_prediction

EDUCATION

Bachelor's Degree in Electronics and Computer Engineering
Rizvi College of Engineering (2023–2027)
Semester 6 | CGPA: 7.27